# POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks:100
Course: Instrumentation | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Assume suitable data if necessary.*

***Attempt all the questions.***

| | | | |
|---|---|---|---|
| 1. | a) | Mention role of the various equipment used in the block diagram of instrumentation system. Along with that demonstrate block diagram with any one instrumentation system. | 7 |
| | b) | Explain in detail about Hay's bridge and mention its application. | 8 |
| 2. | a) | Which bridge circuit will you prefer for the measurement of small unknown resistance of a resistor? Explain with the help of circuit diagram. | 7<br>8 |
| | b) | An ac bridge has in arm AB a pure capacitance of 0.2μF; in arm BC, a pure resistance of 500Ω; in arm CD, a series combination of R=50Ω and L=0.1H. Arm DA consists of a capacitor C=0.4μF in series with variable resistor $R_s$. If the bridge is balanced at 1kHz, find the value of $R_s$ to obtain bridge balance | |
| 3. | a) | Explain the principle of operation of LVDT. List out its area of applications. | 8 |
| | b) | Define piezoelectric effect. Derive the expression for output voltage in a piezoelectric transducer when stress is applied. | 7 |
| 4. | a) | Explain about strain gauge and show relation between Poission's ratio and $G_f$. | 8 |
| | b) | Explain the working principle of instrumentation amplifier. | 7 |
| 5. | a) | What digital output you will find of 9.27 volts input from 5-bit SA ADC with reference voltage of 15V. | 8 |
| | b) | Define data transmission. Why is it necessary in measurement system? Explain the different modes of data transmission. | 7 |
| 6. | a) | Explain the principle of operation of X-Y recorder. List out its area of application. | 7 |
| | b) | Briefly explain about the digital data acquisition system. | 8 |

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7. Write short notes on: (**Any two**) 2×5

a) Op-amp as a inverter

b) Static performance characteristics of measurement system

c) Strip-chart recorder

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